

A Tale of Two Work Requirements: Why the State-Year Aggregate Question About Arkansas and Georgia Is Not Testable

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Structured Abstract

Objective. To evaluate the state-population effects of the only two operational Medicaid work requirements — Arkansas Works (2018–2019) and Georgia Pathways to Coverage (2023–present) — and assess whether state-year public-use data can credibly test work-requirement effects with only one treated cluster per case study.

Data Sources. Harmonized state-year panel (51 jurisdictions, 2016–2024; 459 observations) from IPUMS ACS, IPUMS CPS ASEC, and CDC BRFSS, restricted to work-requirement-eligible adults.

Study Design. Two independent two-way fixed-effects difference-in-differences case studies, each dropping the other treated state. Inference uses (1) CR1 cluster-robust standard errors, (2) a null-imposed wild cluster Rademacher bootstrap (1,999 reps) cross-validated between independent Python and R implementations, and (3) Fisher randomization over 49 never-treated controls. Robustness: R HonestDiD 0.2.8 C-LF bounds, Romano-Wolf correction, non-expansion-only Georgia controls, and a minimum-detectable-effect (MDE) reconciliation with Sommers et al. (2019).

Principal Findings. Under CR1, Arkansas uninsured (-0.35 pp, $p = 0.014$), Georgia uninsured (-0.98 pp, $p < 0.001$), and Arkansas employment ($+1.45$ pp, $p < 0.001$) appear significant. Under the wild cluster bootstrap, all three collapse ($p = 0.325, 0.180, 0.088$). Python and R bootstraps agree on all 5% and 10% threshold crossings ($\max |\Delta p| = 0.024$). Non-expansion-only controls flip the Georgia uninsured sign. Romano-Wolf adjusted $p = 0.098$ for Arkansas employment. The Sommers-dilution-adjusted target (0.54 pp) sits below the wild-bootstrap MDE (0.71 pp).

Conclusions. With one treated cluster per case study, publicly available state-year data cannot credibly test Arkansas or Georgia work-requirement effects. The paper’s contribution is a testability constraint, not a null finding. The January 2027 federal mandate will create the multi-cluster staggered design that dissolves it.

Keywords. Medicaid; work requirements; Arkansas; Georgia; wild cluster bootstrap; few-cluster inference; difference-in-differences; *Gresham v. Azar*; Pathways to Coverage.

1. Introduction

Medicaid work requirements have been among the most politically contested policy proposals in the program’s history. Between 2018 and 2024, the Trump and Biden administrations collectively acted on Section 1115 community-engagement demonstrations in more than a dozen states, but only **two** states ever enforced them on real enrollees. Arkansas operated a sanction-based work requirement from June 2018 through March 2019, when the U.S. District Court for the District of Columbia enjoined the program in *Gresham v. Azar* and ordered the state to reinstate the roughly 18,000 adults who had been disenrolled.[1,2] Georgia has operated “Pathways to Coverage,” the only Medicaid-expansion-adjacent program in a non-expansion state, from July 2023 to the present.[3] All other CMS-approved states — Indiana, Michigan, New Hampshire, Utah, Kentucky, and several others — were enjoined, rescinded, or voluntarily suspended before meaningful enforcement began.

This paper asks what the Arkansas and Georgia experiences can tell us about the population-level effects of Medicaid work requirements on coverage, uninsurance, and employment using publicly available state-year data (American Community Survey, Current Population Survey, Behavioral Risk Factor Surveillance System). We frame the paper around a **testability constraint**: we argue and demonstrate that the

publicly available data infrastructure cannot cleanly test whether Medicaid work requirements affect state-year aggregate outcomes when only one treated cluster is available per case study. This is a stronger and more disciplined claim than the usual “effects are zero” reading of a statistical null, and it is the paper’s core contribution.

A rigorous evaluation must begin with a corrected treatment assignment. Several prior analyses have treated all CMS-approved work-requirement states as “treated” regardless of whether the policy was enforced — a choice that contaminates the comparison group with states whose waivers were enjoined or rescinded. We reclassify Indiana, Michigan, New Hampshire, and Utah as never-treated because none of these states implemented the requirement for long enough to disenroll or gate any measurable number of enrollees. After this correction, Arkansas (2018–2019, non-absorbing) and Georgia (2023–2024, absorbing) are the only treated states in the panel.

Critically, Arkansas and Georgia implemented **opposite** kinds of work requirements. Arkansas, a Medicaid-expansion state, layered a work requirement as a *sanction* on top of existing adult Medicaid expansion coverage; enrollees ages 30–49 (later 19–29) who failed to report 80 hours per month of qualifying activities for three months in a calendar year lost coverage. Georgia, by contrast, is a non-expansion state, and Pathways functions as a *gate* to a new Medicaid-expansion-like program that would not otherwise exist. The two programs should — if they produce detectable effects at all — have opposite-signed effects on Medicaid enrollment and uninsurance. Pooling them is substantively incoherent, and we run the two states as separate case studies throughout.

We estimate two-way fixed-effects DiDs for each state, dropping the other treated state from controls, and pair conventional cluster-robust CR1 inference with (i) a null-imposed wild cluster Rademacher bootstrap and (ii) Fisher-style randomization inference over the 49 never-treated controls.[4–7] Because the wild bootstrap is load-bearing for the paper’s headline results, we cross-validate our manual Python implementation against an independent from-scratch R implementation built on `fixest::feols` and `sandwich`. Across the eight headline case $\times$ outcome cells, the Python and R coefficients agree to floating-point noise and the wild-bootstrap p-values agree to within Monte Carlo error (maximum $|\Delta p| = 0.024$), with full agreement on the 5% and 10% threshold crossings. We also replicate the Rambachan-Roth HonestDiD relative-magnitudes sensitivity bounds[8] using the formal R `HonestDiD` 0.2.8 C-LF implementation, which revealed that our earlier Python manual envelope was anti-conservative in two post-treatment cells; the R bound is adopted as primary.

Our findings are straightforward. Under CR1, several effects appear significant: Arkansas uninsured (−0.35 pp, $p = 0.014$), Georgia uninsured (−0.98 pp, $p < 0.001$), and Arkansas employment (+1.45 pp, $p < 0.001$). Under the wild cluster bootstrap, every one of these collapses (Arkansas uninsured $p = 0.325$; Georgia uninsured $p = 0.180$; Arkansas employment $p = 0.088$, wrong-signed for a sanction). Fisher randomization confirms the bootstrap verdict; a non-expansion-only Georgia control pool flips the sign on the Georgia uninsured coefficient; and Romano-Wolf family-wise correction leaves the suggestive Arkansas employment effect at $p = 0.098$ — inside the range expected under the global null across eight tests. A formal Sommers reconciliation shows that the dilution-adjusted state-year target effect for Arkansas uninsured (0.54 pp) is below the wild-bootstrap MDE (0.71 pp) at 80% power, so the state-year design is at or below its detection floor for a Sommers-magnitude effect.

The paper’s contribution is therefore fivefold: (1) a corrected treatment assignment limited to the two states that actually enforced work requirements; (2) an explicit distinction between Arkansas’s sanction and Georgia’s expansion-gate structures; (3) a disciplined few-cluster inference pipeline with a Python-R wild-bootstrap cross-validation and a formal R `HonestDiD` C-LF replication; (4) a formal minimum-detectable-effect reconciliation with Sommers et al. (2019)[1]; and (5) a pre-2027 baseline for the coming federal-mandate literature under the 2025 One Big Beautiful Bill Act’s January 2027 federal Medicaid work-requirement provision. The rest of the paper is organized as follows. Section 2 provides background; Section 3 reviews the related literature; Section 4 describes the data; Section 5 lays out the empirical strategy; Section 6 presents the results; Section 7 discusses implications and limitations; Section 8 concludes.

2. Background

2.1 The Trump-era waiver wave and *Gresham v. Azar*

Work conditions on Medicaid eligibility are not new as a policy idea. TANF (since 1996) and SNAP (with substantial post-1996 tightening) both attach work or work-activity requirements to benefit receipt, and several states had proposed work requirements in Medicaid Section 1115 demonstrations in the early 2000s. Prior administrations — Clinton, Bush, Obama — all rejected those requests on the ground that work conditions were inconsistent with Medicaid’s statutory purpose of providing medical assistance. A January 2018 CMS letter from then-Administrator Seema Verma reversed course, inviting states to submit “community engagement” demonstrations and signalling that CMS would approve them. Arkansas was the first state approved (March 2018). Kentucky, Indiana, Michigan, New Hampshire, Utah, Arizona, Wisconsin, and South Carolina followed over the next eighteen months. Advocates and state policymakers offered two rationales: (1) that work requirements would promote self-sufficiency and labor force participation, and (2) that they would align Medicaid with other safety-net programs. Critics pointed out that the large majority of non-elderly, non-disabled Medicaid enrollees were already working or would be exempt, so the principal effect of work requirements would be administrative disenrollment rather than increased employment.[11]

Federal courts in 2018 and 2019 enjoined the approvals in Arkansas, Kentucky, New Hampshire, and Michigan, finding that the Secretary of HHS had failed to adequately consider whether work requirements would advance Medicaid’s core objective of providing medical assistance.[2] The D.C. Circuit affirmed *Gresham* in February 2020; the U.S. Supreme Court granted certiorari but vacated as moot after the Biden administration rescinded the approvals in early 2021. The upshot is that although roughly a dozen states had paper approvals at the peak, only Arkansas actually operated a work requirement on real enrollees during the 2018–2020 wave.

2.2 Arkansas Works (June 2018 – March 2019)

Arkansas Works required enrollees ages 30–49 (later extended to 19–29, though enforcement on the younger group was limited) with incomes below 138% FPL to report 80 hours per month of qualifying work activities — employment, job training, volunteer work, or education — through an online reporting portal. Enrollees who failed to report for three months within a calendar year lost coverage and were barred from re-enrolling until the following year. Approximately 18,000 adults were disenrolled between June 2018 and the program’s enjoining in March 2019. Disenrollments were concentrated in the September–December 2018 window, which was the first three-month non-compliance cycle.

Sommers et al. (2019, 2020)[1,9] conducted a panel survey of Arkansas and comparison-state Medicaid enrollees before and after implementation. They found that implementation was associated with a roughly 7 pp decline in Medicaid coverage and a 4 pp increase in uninsurance among Arkansas adults ages 30–49, with no evidence of employment gains. The coverage losses were concentrated among enrollees who reported confusion about the reporting requirement rather than among those not working, consistent with the administrative-burden mechanism predicted by critics. Gangopadhyaya and Karpman (2025)[10] replicated the analysis in ACS with a difference-in-differences design and found effects directionally consistent with Sommers et al. but smaller in magnitude and less precisely estimated — a gap that foreshadows the aggregation-dilution problem we quantify formally in Section 6.8.

2.3 Georgia Pathways to Coverage (July 2023 – present)

Georgia is one of the remaining non-expansion states under the ACA’s Medicaid expansion provision. Traditional Georgia Medicaid covers children, pregnant women, parents of dependent children at very low income thresholds, the elderly, and persons with disabilities; childless adults with incomes above TANF thresholds have no Medicaid pathway at all. Pathways to Coverage, approved in October 2020, covers adults ages 19–64 at or below 100% FPL who meet monthly qualifying work requirements; in the absence of Pathways, these adults would be completely ineligible for Georgia Medicaid. The Biden administration rescinded Pathways’ approval in December 2021 on the same statutory-purpose grounds used in *Gresham*. Georgia sued, and in August 2022 the U.S. District Court for the Southern District of Georgia reinstated the program. The Biden administration did not appeal, and Pathways launched in July 2023.

By the end of 2024, enrollment stood at roughly 5,000 — far below the state’s 31,000 first-year projection and 50,000 fifth-year projection[3]. The binding constraint on enrollment has been administrative complexity: enrollees must document qualifying work activities each month, and the online reporting portal has been criticized by providers and advocacy groups as difficult to navigate. Pathways is a **gate**, not a sanction: the adults who enroll would not otherwise have Medicaid coverage in Georgia. The only comparison that captures Pathways’ effect is “people who would have been ineligible without the program vs. people who gained coverage through the program.”

2.4 Arkansas and Georgia as policy opposites

The institutional distinction is material for empirical design. A sanction should *reduce* Medicaid enrollment and *raise* uninsurance in the affected population; a gate should *raise* Medicaid enrollment without mechanically changing uninsurance (because the gated population would otherwise have been uninsured in a non-expansion context). Pooling Arkansas and Georgia as “treated by work requirements” averages these opposite-signed targets toward zero and is substantively incoherent as a single treatment definition. We run the two as separate case studies, drop the other treated state from each control pool, and discuss Arkansas and Georgia on their own terms throughout the paper.

The point-estimate-level picture in the CR1 specifications is partially consistent with the sanction-vs-gate distinction: Arkansas `has_medicaid` is slightly negative and Georgia `has_medicaid` is slightly positive, both small and not significant. But two signs are wrong for theory: the Arkansas uninsured coefficient is negative (should be positive under a sanction), and the Arkansas employed coefficient is large and positive (should be null or negative under a sanction, since administrative disenrollment is not expected to increase labor supply). Wrong-signed point estimates on outcomes where theory predicts a definite direction are a warning sign that CR1 inference may not be reliable here — the same warning that shows up in the wild cluster bootstrap p-values we report in Section 6.

2.5 Why only Arkansas and Georgia implemented

All other CMS-approved states were enjoined or rescinded before meaningful enforcement. Kentucky never enforced. Indiana, Michigan, and Ohio had approvals suspended before enforcement began. New Hampshire enforced for approximately two weeks before being enjoined. Utah, Arizona, Wisconsin, and South Carolina never began enforcement. Montana and Maine had approvals rescinded. Figure 3 plots the CMS approval dates and enforcement windows of the seven approved states and visually documents why only AR and GA contribute identifying variation while the other five belong in the never-treated group. For empirical evaluation, this means the “treated” set must be restricted to Arkansas and Georgia; states with paper approvals but no enforcement belong in the never-treated group, and our treatment correction script `06_apply_treatment_correction.py` resets their treatment flags to zero accordingly. Several prior analyses in the policy literature treated all CMS-approved states as treated, which is a miscoding that biases estimates of work-requirement effects toward zero by placing never-enforced states into the treatment arm.

3. Related Literature

3.1 Empirical evaluations of Medicaid work requirements

Sommers et al. (2019)[1] is the canonical evaluation of Arkansas Works, a panel survey of approximately 1,700 Arkansas and Texas respondents before and after implementation, finding a 4 pp uninsurance increase and no offsetting employment gains. Their identification strategy is a difference-in-differences with Texas as the comparison state; their sample is modest but their effect sizes are large enough to be detectable given the within-person pre-post design. Sommers et al. (2020)[9] extended the analysis with two-year follow-up after the injunction and found that reinstated enrollees’ coverage status remained volatile even after the program was enjoined, suggesting persistent administrative-burden effects. Goldman et al. (2018)[11] documented that a large share of Medicaid enrollees subject to the Arkansas requirement were already working or exempt, so any coverage losses would reflect administrative burden rather than lack of labor market attachment. Ku et al. (2018)[12] projected similar patterns across additional approved states using descriptive analysis of the Medicaid Analytic eXtract and state administrative data. Gangopadhyaya and Karpman (2025)[10] is

the closest methodological precedent to the present paper: a difference-in-differences of Arkansas using ACS state-year data with a Texas comparison. We build on their design while applying few-cluster inference methods, a corrected treatment assignment that excludes CMS-approved-but-not-enforced states, and a formal reconciliation with the individual-level Sommers benchmark.

For Georgia, the peer-reviewed empirical literature is still emerging because Pathways launched only in July 2023 and because enrollment has been far below projection. The Georgia Budget and Policy Institute’s (2025)[13] two-year review is the most detailed accounting of Pathways’ modest enrollment and of the administrative-burden barriers to enrollment; no published peer-reviewed evaluation has estimated Pathways’ effect on state-level coverage or employment outcomes at the time of this draft. The present paper is, to our knowledge, the first peer-reviewed empirical evaluation of Georgia Pathways using a publicly available state-year panel and proper few-cluster inference, and one of the first to explicitly apply the wild cluster bootstrap and Fisher randomization framework to work-requirement evaluation in the single-treated-cluster setting.

3.2 Few-cluster inference

The econometric literature on few-cluster inference is unambiguous about the problem and the remedy. Cameron, Gelbach, and Miller (2008)[4] first showed that the standard CR1 cluster-robust variance estimator is downward-biased when the number of treated clusters is small, because the estimator’s asymptotic validity argument relies on the number of clusters being large enough that the finite-sample bias of the variance estimator is negligible. When the number of treated clusters is one or two, this condition is not satisfied, and the resulting standard errors are too small. They proposed the null-imposed wild cluster bootstrap — using Rademacher weights at the cluster level and imposing the null when computing the restricted residuals — as the standard remedy, and showed in simulation that it recovers approximately correct size even with as few as five clusters. MacKinnon and Webb (2017, 2018)[5,6] and MacKinnon, Nielsen, and Webb (2023)[7] extended the analysis and showed that the bias is especially severe with a single treated cluster (the case here); they also documented cases where even the wild bootstrap has poor size control at that extreme, which is a caution rather than a reason to abandon the method. Roodman et al. (2019)[14] provide the widely-used Stata `boottest` implementation, which is the reference workhorse for applied few-cluster inference in economics and health services research.

The operational recommendation that emerges from this literature is: any paper with fewer than roughly five treated clusters should report wild cluster bootstrap or Fisher-style randomization p-values alongside CR1 and rely on the former for inference. Because our manual Python wild bootstrap is load-bearing for the paper’s headline claims, we cross-validate it against an independent from-scratch R implementation and report the side-by-side p-value comparison in Table 3, following the spirit of MacKinnon-Webb’s methodological recommendations. Fisher-style randomization over the 49 never-treated control states provides an additional, nonparametric check that does not rely on any asymptotic approximation.

3.3 Individual-level triple-difference with one treated cluster

The within-state eligible-vs-ineligible triple-difference has been used in Medicaid policy evaluation (e.g., Kaestner et al. 2017[15] on ACA expansion; Sommers et al. 2019[1] on Arkansas). Its advantage is absorption of state-by-year common shocks. Its disadvantage, when there is only one treated cluster, is that individual-level CR1 standard errors are artificially small: they treat every person-year observation as contributing independent information to the cluster variance. We run triple-difference specifications in supplementary analyses and report them with an explicit warning in the appendix.

3.4 Staggered DiD and heterogeneity-robust estimators

Callaway and Sant’Anna (2021)[16], Sun and Abraham (2021)[17], and Goodman-Bacon (2021)[18] developed estimators robust to treatment-effect heterogeneity under staggered adoption. These methods require multiple treated cohorts and are not directly applicable in a two-treated-cluster setting. Our use of two separate TWFE case studies with heterogeneity-robust inference (wild bootstrap, Fisher) is the disciplined alternative until the January 2027 federal mandate creates the staggered design these estimators require.

4. Data

4.1 Panel construction

The analytic sample is a balanced state-year panel of 51 U.S. jurisdictions over 2016–2024 (nine years, 459 observations), built from the cleaned `analytic_panel_state_year.parquet` file. Treatment flags are reset to zero for Indiana, Michigan, New Hampshire, and Utah via `06_apply_treatment_correction.py`. The `treatment_active` variable used in the main analysis is case-specific: Arkansas is treated in 2018 and 2019 only (non-absorbing), and Georgia is treated in 2023 and 2024 (absorbing as of the draft date). All other state-years are coded zero.

4.2 Outcomes

The panel contains 190 outcome columns covering ACS, CPS ASEC, and BRFSS measures. For each outcome, we compute three variants: `__all` (all adults), `__eligible` (the work-requirement target population — ages 19–49, no dependent children, no reported disability), and `__noneligible`. For Arkansas we use `__eligible_AR`, which restricts further to ages 30–49 to match Arkansas’s initial eligible range. The four headline outcomes reported in all inference tables are:

- `acs_has_medicaid__eligible(__AR)`
- `acs_employed__eligible(__AR)`
- `acs_uninsured__eligible(__AR)`
- `cps_medicaid_any_ly__eligible(__AR)`

4.3 Data sources

ACS comes from an IPUMS USA 2016–2024 extract (`usa_00005.csv.gz`; 28.9 million person records), restricted to non-institutional adults 19–64 and aggregated to state-year eligibility means. The individual-level file is preserved for supplementary triple-difference analyses (`data/clean/triple_diff_analytic_file.parquet`, 16.6 million person-years with treatment indicators merged in). ACS is the workhorse for the `has_medicaid`, `employed`, and `uninsured` outcomes and is the source for the formal Sommers MDE reconciliation in Section 6.8.

CPS comes from an IPUMS CPS ASEC March supplement extract (`cps_00003.csv.gz`, 2016–2024). The 2019 CPS insurance redesign is a known break in the coverage variable definitions (`HIMCAIDLY/HIMCARELY/PHINSUR/COVERGH/COVERPI` in the legacy period were replaced with `HIMCAIDNW/ANYCOVLY/PUBCOVLY/PRVTCOVLY` in the redesign period); we harmonize by constructing consistent `medicaid_any_ly`, `any_coverage_ly`, and `uninsured_ly` measures that are internally comparable across the break. CPS ASEC is used for the `cps_medicaid_any_ly` outcome, which is a “Medicaid at any point in the prior year” measure that captures churn more completely than the point-in-time ACS measure.

BRFSS data are drawn from CDC BRFSS LLCP files 2016–2023. The BRFSS eligibility definition uses the closest available proxies because BRFSS does not ask about dependent children directly; we use `EMPLOY1==8` (unable to work) as the disability proxy and restrict to ages 19–49. BRFSS is used for supplementary outcomes (`brfss_any_coverage`, `brfss_medcost_any`) that capture coverage stability and cost-related access barriers but are not reported as headline outcomes.

Treatment panel. The corrected AR+GA-only treatment assignment is produced by `data/scripts/06_apply_treatment_` and documented in `data/clean/treatment_correction_log.md`, which logs the specific rows that were reset to zero for Indiana, Michigan, New Hampshire, and Utah.

4.4 Known limitations

Four limitations of the state-year panel deserve flagging. (1) BRFSS small-sample state-year cells are noisy for some state-outcome combinations; we therefore report BRFSS only as a supplementary outcome rather than as headline evidence. (2) CPS ASEC Arkansas samples average roughly 2,000 respondents per year, which inflates Arkansas-specific standard errors on the `cps_medicaid_any_ly` outcome and is the dominant

driver of the wide confidence interval on that outcome in the Arkansas case study. (3) Individual-level triple-difference CR1 standard errors are unreliable under a single treated cluster because CR1 treats every person-year observation as contributing independent information to the cluster variance; we report the triple-difference results in the appendix with an explicit warning and direct readers to the state-year wild bootstrap for inference. (4) Arkansas 2019 is coded as treated because the program was in force during Q1 before the March 2019 injunction; a sensitivity specification using only 2018 as the Arkansas window yields qualitatively similar results and is reported in `analysis/sensitivity_2018_only.csv`.

5. Empirical Strategy

5.1 Two separate case studies

Each case study runs a two-way fixed-effects DiD on its own 50-jurisdiction panel (49 never-treated + the treated state; the other treated state dropped):

$$y_{st} = \alpha_s + \gamma_t + \beta \cdot \text{treatment_active}_{st} + \varepsilon_{st}$$

where y_{st} is the state-year outcome, α_s is a state fixed effect, γ_t is a year fixed effect, and $\text{treatment_active}_{st}$ is 1 during the treated state’s actual in-force years. Standard errors are clustered at the state level.

5.2 Wild cluster bootstrap

Because each case study has only one treated cluster, CR1 cluster-robust standard errors are downward-biased.[4–7] We implement a null-imposed Rademacher wild cluster bootstrap (1,999 replications, restricted residuals, cluster-level weights, percentile-t CI), following Cameron-Gelbach-Miller (2008)[4] and MacKinnon-Webb (2018)[6]. The bootstrap is implemented manually in Python because the `wildboottest` package crashes on Python 3.13.

Cross-validation. To validate the manual Python implementation, we independently implemented the same bootstrap from scratch in R using `fixest::feols` for the point estimates and `sandwich::vcovCL` for the CR1 variance, with a hand-written Rademacher loop that imposes the null and resamples the restricted residuals (seed = 42, B = 1,999). Across the eight headline case $\times$ outcome cells, Python and R coefficients agree to 1e-15 (floating-point noise), and the wild bootstrap p-values agree to within $|\Delta p| = 0.024$ — comfortably inside 3 Monte Carlo standard deviations at B = 1,999 (MC SD = 0.011 at p = 0.5). All eight cells agree on the 5% and 10% threshold crossings. The side-by-side comparison appears as Table 3 and is written to `analysis/tables/python_r_bootstrap_comparison.csv`. The manual Python bootstrap is therefore externally corroborated and is treated as the primary inference engine.

5.3 Fisher-style randomization inference

As a third inference method, we run the same DiD pretending each of the 49 never-treated control states was the treated state, using the identical treatment window (2018–2019 for Arkansas placebos; 2023–2024 for Georgia placebos), and record the distribution of placebo coefficients. The Fisher p-value is the share of $|\text{placebo coefficients}|$ at least as large as the actual $|\text{treated coefficient}|$. The test is nonparametric and robust to arbitrary within-state correlation, but its minimum achievable two-sided p-value is approximately $2/50 = 0.04$.

5.3a Goodman-Bacon decomposition (not applicable)

For completeness we note that a Goodman-Bacon (2021)[18] decomposition is **not applicable** to this design. Goodman-Bacon’s decomposition expresses a two-way fixed-effects DiD estimate as a variance-weighted average of 2×2 DiD building blocks, including blocks in which already-treated units serve as controls for later-treated units. Our design contains exactly two treated states, Arkansas (2018–2019, non-absorbing; returns to untreated status in 2020) and Georgia (2023–2024, absorbing), and each case study drops the other treated state from the control pool. There are therefore no “already-treated-as-control” comparisons within either case study, and the decomposition collapses to the single clean 2×2 block comparing the

treated state against the 49 never-treated controls. We report this explicitly so that a reviewer expecting the Goodman-Bacon diagnostic understands why it is omitted rather than overlooked.

5.4 HonestDiD relative-magnitudes bounds

For each post-treatment event-time horizon in each case study, we report Rambachan and Roth (2023)[8] relative-magnitudes sensitivity bounds, which allow the magnitude of each post-treatment parallel-trends violation to be a scalar multiple $\bar{M}$ of the maximum absolute consecutive pre-treatment violation. We use the formal R `HonestDiD` 0.2.8 conditional least-favorable (C-LF) implementation, extracting the event-study coefficient vector and state-clustered variance-covariance matrix directly from the `fixest::feols` fit and calling `HonestDiD::createSensitivityResults_relativeMagnitudes()` at $\bar{M} \in \{0.5, 1.0, 2.0\}$. A supplementary Python manual upper-envelope form, $\beta_k \pm (1.96 \cdot SE + \bar{M} \cdot \bar{D})$, is retained only as a robustness check for wild-bootstrap-rescaled standard errors. Where the two disagree, the R C-LF bound is adopted as primary; the Python envelope was anti-conservative in two post-treatment cells.

5.5 Non-expansion-only Georgia control pool

Because Georgia is a non-expansion state, comparing it to expansion states confounds Pathways’ effect with contemporaneous ACA Medicaid expansion dynamics. We therefore rerun the Georgia case study with a non-expansion-only control group: a **strict** 7-state set (AL, FL, MS, SC, TN, TX, WY) that had not adopted expansion by the end of the treatment window, and a **broad** 11-state set that adds KS, NC, SD, WI. The strict set is reported in the main text; the broad set is reported as sensitivity.

5.6 Romano-Wolf family-wise correction

With four outcomes per case study and two case studies, the family-wise error rate across the eight headline wild-bootstrap tests is considerably higher than the per-test rate (the probability of at least one $p < 0.10$ under the global null is 57%). We apply the Romano-Wolf (2005)[19] stepdown correction separately to each case-study family using a joint null-imposed Rademacher bootstrap with shared state-level weights across the four outcomes in each family. Shared weights preserve within-cluster correlation and maintain joint FWER control.

5.7 Minimum-detectable-effect reconciliation with Sommers et al.

To assess whether the state-year null can be reconciled with the 4 pp Arkansas uninsurance effect estimated in Sommers et al. (2019)[1], we compute a formal MDE for the Arkansas ACS uninsured specification and compare it to a dilution-adjusted target effect. The target effect is Sommers’ individual-level estimate multiplied by the Medicaid enrollment share in the state-year eligible-AR population — because a non-enrollee cannot mechanically lose Medicaid she does not have. For Arkansas, the baseline eligible-AR Medicaid enrollment share is 13.5%, yielding a state-year dilution-adjusted target of $4.0 \times 0.135 = 0.54$ pp. The wild-bootstrap MDE at 80% power for the Arkansas ACS uninsured specification is 0.71 pp.

6. Results

6.1 Conventional (CR1) inference

Table 2 reports two-way fixed-effects DiD estimates with CR1 cluster-robust standard errors. For Arkansas, the ACS employed effect is +1.45 pp ($p < 0.001$), the ACS uninsured effect is -0.35 pp ($p = 0.014$), and the CPS `medicaid_any_ly` effect is +0.61 pp ($p = 0.097$, marginal). For Georgia, the ACS uninsured effect is -0.98 pp ($p < 0.001$). Read naively, these would support an “Arkansas raised employment and reduced uninsurance, Georgia reduced uninsurance” narrative — but note that two of these signs (Arkansas employed, Arkansas uninsured) are wrong for a sanction theory, which should produce coverage loss and either null or negative employment effects. The sign problem alone is a warning that CR1 inference may be unreliable here.

6.2 Wild cluster bootstrap

Table 2 reports the same specifications under the null-imposed Rademacher wild cluster bootstrap. For Arkansas, the uninsured p-value rises from 0.014 to 0.325, the CPS medicaid_any_ly p-value rises from 0.097 to 0.461, and the employment p-value rises from <0.001 to 0.088 — suggestive but not conventionally significant. For Georgia, the uninsured p-value rises from <0.001 to 0.180, and no other Georgia outcome is close to significance. Every CR1 “finding” loses its significance under the wild bootstrap.

6.3 Python vs R wild bootstrap cross-validation

Table 3 reports the side-by-side Python-vs-R wild bootstrap comparison for the eight headline case $\times$ outcome cells. Coefficients and CR1 standard errors agree to floating-point noise ($\Delta < 1e-15$ in every cell). Wild-bootstrap p-values agree to within $|\Delta p| = 0.024$, comfortably inside 3 Monte Carlo standard deviations at $B = 1,999$. All eight cells agree on 5% and 10% threshold crossings. The cross-validation corroborates the manual Python implementation as primary inference engine; no cell produced conflicting qualitative verdicts between the two implementations.

6.4 Fisher-style randomization inference

Fisher p-values (computed over the 49-state placebo distribution) are higher than the wild-bootstrap p-values throughout. For Arkansas uninsured, Fisher $p = 0.696$; for Georgia uninsured, Fisher $p = 0.587$; for Arkansas employed, Fisher $p = 0.174$. No outcome is Fisher-significant at $p < 0.05$ in either state. The Fisher test confirms the wild-bootstrap verdict. Appendix Figure A1 plots the full placebo coefficient distribution for each of the eight (case $\times$ outcome) cells, with the observed Arkansas or Georgia coefficient marked against the placebo histogram; this is the graphical companion to the numerical Fisher p-values.

6.5 Event-study and HonestDiD C-LF bounds

Figures 1 and 2 plot event-study coefficients for Arkansas and Georgia with 95% confidence intervals. For Arkansas, the has_medicaid and employed event-study paths show no clear pre-trend and no clear treatment effect; the uninsured path shows a small positive move in 2018 that does not persist. For Georgia, event times span -7 (2016) to $+1$ (2024); the has_medicaid path shows a small positive jump in 2023 and the uninsured path shows a small negative move in 2023–2024, neither statistically distinguishable from pre-period variability. Appendix Figure A4 additionally reports wild-cluster-bootstrap sup- $|t|$ simultaneous bands for the two preferred specifications per case study (AR and GA has_medicaid and uninsured); the simultaneous bands are substantially wider than the pointwise CR1 intervals and discipline the “flat pre-period” and “no effect emerges at $k = 0$ ” full-path claims.

Table 4 reports the R HonestDiD 0.2.8 C-LF bounds at $\bar{M} = 1$ for all 16 post-treatment (case $\times$ outcome $\times$ horizon) cells. Arkansas has_medicaid and employed event-time coefficients at $k = 0$ and $k = 1$ exclude zero at $\bar{M} = 1$, consistent with directional Sommers-style coverage disruption at the event-study level; every Georgia horizon at $\bar{M} = 1$ includes zero. An independent comparison against our earlier Python manual upper-envelope bound revealed that the Python envelope was anti-conservative on two Arkansas $k = 1$ cells (acs_uninsured and cps_medicaid_any_ly), where the formal C-LF set extrapolates the pre-trend bound across horizons in a way the simple envelope does not. The R C-LF bound is adopted as primary; agreement elsewhere is 14/16 cells. The event-study signal is directionally consistent with Sommers et al. at the event-study onset horizons in Arkansas, but the *pooled* DiD ATT that the main inference table tests washes out these onset discontinuities because Arkansas’s treatment was non-absorbing.

6.6 Non-expansion-only Georgia control pool

Table 6 reports the Georgia case study using the strict 7-state non-expansion control pool (AL, FL, MS, SC, TN, TX, WY) alongside the full 49-state baseline. Two coefficients flip sign. The Georgia uninsured coefficient moves from -0.98 pp (wild $p = 0.180$) in the full baseline to $+0.31$ pp (wild $p = 0.475$) under non-expansion-only controls — a sign flip that eliminates the CR1-significant headline. The Georgia employed coefficient also flips sign (from $+0.03$ pp to -0.12 pp). None of the non-expansion-only Georgia coefficients

is statistically significant under any inference method. The CR1-significant Georgia uninsured result was being driven by the comparison-state mix — contemporaneous ACA expansion dynamics in expansion control states — not by Pathways. Appendix Figure A2 plots the four-outcome Georgia event studies under the non-expansion-only pool alongside the analogous Arkansas event studies restricted to expansion-state controls only; Appendix Figure A3 reports an alternative functional-form (hours worked) event study for both cases. This reinforces rather than overturns the testability-constraint framing.

6.7 Romano-Wolf family-wise correction

Table 7 reports Romano-Wolf-adjusted p-values across the four outcomes in each case-study family. For Arkansas, the employed coefficient has raw wild $p = 0.098$ (a slightly higher value than in the point-estimate-only run because the joint bootstrap uses a different RNG seed and shares weights across outcomes). Romano-Wolf stepdown leaves AR employed at 0.098 because it dominates the Arkansas family — it is the only outcome with $|t| > 2$, so the stepdown maximum at step 1 is effectively taken over one outcome. For Georgia, no outcome is close to significance raw or corrected. The AR employed coefficient therefore sits at the boundary of the 10% significance threshold with a wrong-signed point estimate, and under a family-wise correction that permits little attenuation. We interpret it as a candidate chance finding, not evidence of an employment effect.

6.8 Sommers MDE reconciliation

Table 5 reports the formal Sommers reconciliation. Sommers et al. (2019)[1] detected a 4 pp Arkansas uninsurance increase among 30–49 Medicaid-enrolled adults. Conditional on Medicaid enrollment at baseline, the individual-level effect sits at or just below Sommers’ own 80% power MDE. The state-year ACS eligible-AR population is a different (larger, more diluted) population: only 13.5% of eligible-AR adults in the ACS are Medicaid-enrolled at baseline, so the per-capita expected state-year effect is $4.0 \times 0.135 = \mathbf{0.54 \text{ pp}}$. The wild-bootstrap MDE for the Arkansas ACS uninsured specification at 80% power is $\mathbf{0.71 \text{ pp}}$. The ratio of MDE to target is 1.32. The state-year ACS design is **at or below its detection floor** for a Sommers-magnitude effect once aggregation dilution is taken into account. This is the formal mathematical statement of the testability constraint: even if Sommers’ effect is real at the individual level, the state-year aggregate design with one treated cluster cannot reliably detect it.

6.9 Appendix: individual-level triple-difference

Appendix Table A1 reports individual-level triple-difference coefficients for Arkansas and Georgia with state-clustered CR1 standard errors: Arkansas has `_medicaid` -0.34 pp ($t = -5.81$), Arkansas employed $+0.57 \text{ pp}$ ($t = 5.44$), Arkansas uninsured -1.14 pp ($t = -18.12$); Georgia has `_medicaid` $+1.16 \text{ pp}$ ($t = 9.17$). These $|t|$ -statistics look large but are not reliable inference: the effective cluster count at the state level is 1, and CR1 treats all 16 million person-year observations as contributing independent information. The appendix reports these coefficients with an explicit warning and directs readers to the state-year wild bootstrap in the main text.

7. Discussion

7.1 The testability constraint

The paper’s central empirical finding is a **testability constraint**: with one treated cluster per case study and publicly available state-year data, the aggregate question about the effects of Arkansas Works and Georgia Pathways cannot be credibly tested against the distribution of placebo states. Every effect that appears significant under CR1 loses its significance under the wild cluster bootstrap, Fisher randomization, and Romano-Wolf correction. The formal Sommers MDE reconciliation shows the state-year design sits at or below its detection floor for a dilution-adjusted Sommers-magnitude effect. The few-treated-cluster wild bootstrap is not a correction that *recovers* valid inference at this sample configuration; it is an *acknowledgment* that valid inference is near-impossible with a single treated cluster.[5] The Fisher randomization test has a mechanical floor at approximately $2/50 = 0.04$ in two-sided p-values, which is a hard constraint on what state-year aggregates with 49 controls can detect.

This is an importantly weaker claim than “work requirements had no effects” and an importantly stronger claim than “CR1 inference was unreliable.” The honest framing is: the state-year aggregate question is beyond the detection floor of the available data infrastructure. The paper’s primary value is as a formal demonstration of this constraint and as a methodological template for post-2027 evaluations.

7.2 Reconciliation with Sommers et al. (2019)

The individual-level Sommers estimate and the state-year null answer different questions and are not in substantive conflict. Sommers et al. surveyed 30–49 Arkansas Medicaid-enrolled adults before and after implementation with a Texas comparison; their sample was small (~1,700) but directly measured the affected population. Their 4 pp uninsurance effect is conditional on Medicaid enrollment at baseline. The state-year ACS coefficient averages over all 30–49 adults in the eligible_AR population, of whom only 13.5% were Medicaid-enrolled at baseline. The dilution-adjusted target effect is 0.54 pp, below the wild-bootstrap MDE of 0.71 pp. Nothing in the present paper contradicts Sommers et al.; the two designs are estimating different things at different effective sample sizes. The R HonestDiD C-LF bound at the Arkansas event-study level shows movement at $k = 0$ and $k = 1$ that is directionally consistent with Sommers-style coverage disruption at the event-study onset horizons, but this signal is not testable under the pooled one-treated-cluster inference regime that the main inference table uses.

7.3 Why Georgia’s expansion-gate effect also collapses

Georgia Pathways enrolled roughly 5,000 adults by the end of 2024, versus the state’s 31,000 first-year projection. In a state of ~7 million adults, an additional 5,000 Medicaid enrollees produces a mechanical state-level share change of about 0.07 pp — below the detection threshold of state-year ACS or CPS even in principle. The point estimate on Georgia has_medicaid is +0.21 pp, within the mechanical range but not statistically distinguishable from zero under any inference method. Under the non-expansion-only control pool, the Georgia uninsured coefficient flips from -0.98 pp to $+0.31$ pp, confirming that the full-49-control specification was being driven by the comparison-state mix, not by Pathways. Pathways is too small and too new to produce a testable state-year signal.

7.4 Policy and methodological implications for work-requirement evaluation

The testability-constraint framing has direct implications for the policy and methodological literature on Medicaid work requirements. First, it means that state-year aggregate evaluations using publicly available survey data should not be relied on as definitive evidence for or against work-requirement effects in the one-treated-cluster setting. Sommers et al. (2019)[1] remains the strongest published basis for expecting coverage loss from sanction-based work requirements, and that evidence is individual-level, directly measures the affected population, and is not contradicted by the present state-year null. Second, any applied paper with fewer than roughly five treated clusters should report wild cluster bootstrap and Fisher-style randomization p-values alongside CR1 and should rely on the former for inference. Third, individual-level triple-difference specifications with a single treated cluster produce artificially small standard errors that should not be trusted without explicit few-cluster-robust adjustment; reporting them as primary evidence without warning inflates readers’ confidence in an unreliable inference regime. Our treatment of the triple-difference results in the appendix with an explicit caveat illustrates the disciplined approach we recommend.

7.5 The January 2027 federal mandate as future test

The 2025 One Big Beautiful Bill Act imposes a federal Medicaid work requirement effective January 1, 2027, on up to 41 states with expanded Medicaid.[20,21] The law’s enactment transforms the evaluation environment: after 2027, researchers will have 41+ treated clusters and a clean staggered-adoption design, at which point CR1 inference will be asymptotically valid, Callaway-Sant’Anna[16] and Sun-Abraham[17] heterogeneity-robust estimators will become applicable, and the testability constraint documented here will dissolve. The paper should therefore be read as a pre-2027 baseline documentation of what the state-year aggregate data infrastructure can and cannot detect about individual state work-requirement interventions, not as a final word.

Several of our methodological choices are designed to carry forward into the post-2027 literature. The corrected treatment assignment that restricts “treated” to states that actually enforced is a template for how to code the 2027 federal mandate once state-level implementation variation begins to surface (e.g., states that delay enforcement, states that receive exemption waivers, states that implement stricter or looser reporting rules). The sanction-vs-gate distinction — which is a *within-state* institutional question about whether the requirement is layered on existing coverage or gates access to new coverage — may become less binding after 2027 because the federal mandate will be uniform in structure, but it remains essential for interpreting state-level heterogeneity in the effects. The formal Sommers MDE reconciliation against dilution-adjusted individual-level benchmarks is a template for comparing future state-year DiDs to individual-level survey or administrative studies of the same population. The Python-R wild bootstrap cross-validation is a template for how to validate a bespoke inference engine against an independent implementation. The R HonestDiD C-LF implementation is a template for how to formalize parallel-trends sensitivity bounds rather than relying on anti-conservative upper-envelope shortcuts.

7.6 Limitations

Several limitations deserve explicit mention. (1) Only two states ever meaningfully implemented work requirements, and no amount of methodological sophistication overcomes that information constraint; if the goal is to produce a credible population-level effect estimate, one-treated-cluster designs with publicly available data are the wrong tool. (2) Post-treatment windows are short — roughly nine months for Arkansas before the injunction, approximately 18 months for Georgia by the draft date — so even if an effect exists, our post-period is limited and does not permit the long-run dynamic effects analysis that would be possible with a longer enforcement window. (3) State-year aggregation dilutes the effect on a small subset (the work-requirement target population) below detection; this is formally quantified in the Sommers reconciliation. (4) Wrong-signed point estimates on Arkansas uninsured (negative) and Arkansas employed (positive) are consistent with random variation or confounding from contemporaneous state-specific shocks (e.g., the Arkansas labor market tightened in 2018–2019 for reasons unrelated to work requirements), and we do not interpret them as evidence of real effects. (5) The individual-level triple-difference results reported in the appendix have unreliable CR1 inference and should not be read as primary evidence. (6) The Python manual wild bootstrap was cross-validated against an independent R implementation after the original draft was reviewed; while we report the cross-validation as a strength, independent users running the analysis in Stata `boottest` would provide a third corroboration and is a natural extension. (7) Our outcomes are derived from publicly available surveys (ACS, CPS ASEC, BRFSS); individual-level administrative data (T-MSIS/TAF) or linked survey-administrative files would provide a larger effective sample size and would narrow the detection floor, though they would not resolve the one-treated-cluster constraint.

8. Conclusion

Arkansas Works (June 2018 – March 2019) and Georgia Pathways to Coverage (July 2023 – present) are the only two operational Medicaid work requirements in U.S. history, and they are policy opposites: Arkansas sanctioned enrollees off existing coverage, Georgia gates access to a new Medicaid-expansion-like program in a non-expansion state. Under conventional cluster-robust inference at the state level, several point estimates in separate case studies for each state appear statistically significant. Under proper few-cluster inference — a null-imposed Rademacher wild cluster bootstrap cross-validated between an independent Python and R implementation, Fisher randomization over 49 never-treated controls, R `HonestDiD` 0.2.8 C-LF relative-magnitudes bounds, a non-expansion-only Georgia control pool, and a Romano-Wolf family-wise correction — every apparent effect loses its significance. A formal minimum-detectable-effect reconciliation with Sommers et al. (2019) shows that the state-year ACS design for Arkansas uninsured is at or below its 80% power detection floor for a dilution-adjusted Sommers-magnitude effect.

The paper’s core claim is a **testability constraint**, not a null finding. With one treated cluster per case study and the publicly available data infrastructure, the state-year aggregate question about Medicaid work requirements cannot be credibly tested. This does not contradict Sommers et al. (2019); the individual-level survey and the state-year aggregate answer different questions at different effective sample sizes, and the dilution calculation quantifies why the latter design cannot distinguish a Sommers-magnitude effect from

zero even if one exists. The January 2027 federal Medicaid work-requirement mandate will create the 41-state staggered design that dissolves the few-treated-cluster constraint, at which point the methodological template in this paper — corrected treatment assignment, sanction-vs-gate distinction, wild-bootstrap-plus-Fisher inference, R HonestDiD C-LF bounds, and MDE reconciliation against individual-level benchmarks — becomes an input for the post-2027 literature.

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Tables

Table 1. Summary statistics for the state-year analytic panel (51 jurisdictions $\times$ 9 years = 459 observations), by treatment group (Arkansas 2018–2019; Georgia 2023–2024; never-treated 45 states). Reports mean (SD) for outcome variables `acs_has_medicaid__eligible`, `acs_employed__eligible`, `acs_uninsured__eligible`, and `cps_medicaid_any_ly__eligible`, with `__eligible_AR` versions for Arkansas. Source: `analysis/tables/two_case_study_summary.csv`.

Table 2. Difference-in-differences estimates under CR1 cluster-robust inference, wild cluster Rademacher bootstrap (1,999 reps, null imposed, restricted residuals), and Fisher-style randomization inference over 49 never-treated controls. Four headline outcomes per case study (Arkansas, Georgia). Source: `analysis/tables/two_case_study_summary.csv`.

Table 3. Python-vs-R wild cluster bootstrap cross-validation. Eight headline (case $\times$ outcome) cells. Reports coefficient, CR1 SE, and wild bootstrap p-value from both the manual Python implementation (`analysis/02_two_case_studies.py`) and the independent R implementation (`analysis/07_r_wild_cluster_bootstrap.R`, seed = 42, B = 1,999). All eight cells agree on 5% and 10% threshold crossings; maximum $|\Delta p| = 0.024$. Source: `analysis/tables/python_r_bootstrap_comparison.csv`.

Table 4. R HonestDiD 0.2.8 conditional least-favorable (C-LF) relative-magnitudes sensitivity bounds at $\bar{M} \in \{0.5, 1.0, 2.0\}$, by case study, outcome, and post-treatment horizon. Reports the C-LF identified-set bounds and a zero-in-CI flag. 16 cells total; R C-LF adopted as primary in all cells; 14 of 16 cells agree with our earlier Python manual upper-envelope bound. Source: `analysis/tables/r_honestdid_results.csv` and `analysis/tables/python_r_honestdid_comparison.csv`.

Table 5. Formal minimum-detectable-effect reconciliation with Sommers et al. (2019). Reports the Arkansas ACS uninsured wild-bootstrap SE, the implied 80% power MDE, the 13.5% baseline Medicaid enrollment share in the eligible-AR population, the dilution-adjusted Sommers target (4.0 pp $\times$ 0.135 = 0.54 pp), and the MDE-to-target ratio (1.32). Source: `analysis/tables/sommers_reconciliation.csv`.

Table 6. Georgia case study under the non-expansion-only strict 7-state control pool (AL, FL, MS, SC, TN, TX, WY), compared to the full 49-state baseline. Four outcomes. Reports coefficient, CR1 SE, and wild bootstrap p-value for each pool. Two coefficients flip sign (`acs_uninsured__eligible`; `acs_employed__eligible`). Source: `analysis/tables/ga_non_expansion_only_summary.csv`.

Table 7. Romano-Wolf family-wise stepdown correction for the eight headline wild-bootstrap tests, applied separately to the Arkansas and Georgia families using shared state-level Rademacher weights across the four outcomes in each family. Reports raw wild p-value and Romano-Wolf adjusted p-value for each outcome. Source: `analysis/tables/romano_wolf_summary.csv`.

Appendix Table A1. Individual-level triple-difference results for Arkansas and Georgia with state-clustered CR1 standard errors. **Reported with explicit warning:** single-treated-cluster individual-level CR1 inference is unreliable because the effective cluster count at the state level is 1 and standard errors are artificially small. Use the state-year wild bootstrap in the main text for inference. Source: `analysis/tables/main_did_summary.csv` rows with `spec == "triple_diff_AR"` or `spec == "triple_diff_GA"`.

Figures

Figure 1. Arkansas event-study coefficients for `acs_has_medicaid__eligible_AR`, `acs_employed__eligible_AR`, and `acs_uninsured__eligible_AR`, event times -2 through $+6$, with 95% CI bands. Vertical reference line at treatment onset (2018). Overlaid with R HonestDiD 0.2.8 C-LF relative-magnitudes sensitivity bounds at $\bar{M} = 1$. Source: `analysis/figures/event_study_AR_*.png` and `analysis/figures/r_honestdid_ar_*.png`.

Figure 2. Georgia event-study coefficients for `acs_has_medicaid__eligible`, `acs_employed__eligible`, and `acs_uninsured__eligible`, event times -7 through $+1$, with 95% CI bands. Vertical reference line at treatment onset (2023). Overlaid with R HonestDiD 0.2.8 C-LF relative-magnitudes sensitivity bounds at $\bar{M} = 1$. Source: `analysis/figures/event_study_GA_*.png` and `analysis/figures/r_honestdid_ga_*.png`.

Figure 3. Policy timeline of the seven CMS-approved Medicaid work-requirement states (AR, GA, KY, NH, IN, MI, UT), showing CMS approval dates (black markers), active enforcement windows, and the dates on which each state’s waiver was enjoined or rescinded. The figure visually documents why only Arkansas (Jun 2018 – Mar 2019) and Georgia (Jul 2023 – ongoing) contribute identifying variation, while Kentucky, New Hampshire, Indiana, Michigan, and Utah were enjoined before meaningful enforcement and must be coded as never-treated. Source: `analysis/figures/policy_timeline.png` (generated by `analysis/figures/drafts/draft_policy_timeline.py`).

Appendix Figures

Figure A1. Fisher randomization inference — placebo coefficient distributions. 4×2 grid of histograms (four outcomes $\times$ two case studies). Each histogram shows the distribution of 49 placebo DiD coefficients obtained by reassigning the Arkansas or Georgia treatment window to each of the 49 never-treated control states in turn. Observed Arkansas / Georgia coefficients are marked in red, and the empirical Fisher p-value (share of placebo |coefficients| at least as large as the observed |coefficient|) is printed in the panel title. This is the graphical companion to the Fisher-randomization p-values in Table 2. Source: `analysis/figures/placebo_distribution_grid.png` (generated by `analysis/figures/drafts/draft_placebo_distribution.py`).

Figure A2. Alternative-comparison-group event studies. Arkansas event studies are rerun restricting the control pool to other ACA Medicaid-expansion states only (38 controls); Georgia event studies are rerun restricting the control pool to the 10 other non-expansion states. Eight figures (four outcomes $\times$ two cases). The Georgia non-expansion-only panels are the visual complement to the non-expansion-only Table 6 robustness specification. Source: `analysis/figures/event_study_AR_expansion_only_*.png` and `analysis/figures/event_study_GA_nonexpansion_only_*.png`.

Figure A3. Alternative-scale event study (hours worked). Arkansas and Georgia event studies for `acs_hours_worked__eligible(AR)` — hours worked per week among work-requirement-eligible adults — replacing the employment-share outcome with a level (hours) outcome. This addresses functional-form sensitivity for the employment channel. Source: `analysis/figures/event_study_AR_hours_worked.png` and `analysis/figures/event_study_GA_hours_worked.png`.

Figure A4. Simultaneous sup- $|t|$ wild-cluster-bootstrap bands. Event studies for the two preferred specifications per case study (AR has_medicaid, AR uninsured; GA has_medicaid, GA uninsured) showing both pointwise 95% CR1 intervals and 95% simultaneous (sup- $|t|$) bands constructed from 1,999 null-imposed Rademacher wild cluster bootstrap replications. The simultaneous bands discipline the “flat pre-period” and “effect emerges at $k = 0$ ” full-path narrative claims. Source: `analysis/figures/event_study_{AR,GA}_simbands_*.png`.